

Model discrepancy formulation

Discrepancy parameterization

$$\delta(\mathbf{z}, \boldsymbol{\theta}) = \begin{bmatrix} \mathbf{I}_{n_u} & \mathbf{I}_{n_u} \otimes \mathbf{z}^\top \mathbf{M}_z \end{bmatrix} \boldsymbol{\theta}$$

Discrepancy prior, noise model

$$\begin{aligned} \boldsymbol{\theta}_{\text{prior}} &\sim \mathcal{N}(\mathbf{0}, \mathbf{W}_\theta^{-1}) \\ \boldsymbol{\varepsilon} &\sim \mathcal{N}(\mathbf{0}, \alpha \mathbf{M}_u^{-1}) \end{aligned}$$

Data and surrogate problem information

Large-scale / surrogate data

$$\mathbf{s}(\mathbf{z}_\ell), \tilde{\mathbf{s}}(\mathbf{z}_\ell), \quad \ell = 1, \dots, n_d$$

Solution to surrogate-constrained optimization problem

$$\tilde{\mathbf{z}} = \operatorname{argmin}_{\mathbf{z}} J(\tilde{\mathbf{s}}(\mathbf{z}), \mathbf{z})$$

Bayesian discrepancy calibration

$$\delta(\mathbf{z}_\ell; \boldsymbol{\theta}') \approx \mathbf{s}(\mathbf{z}_\ell) - \tilde{\mathbf{s}}(\mathbf{z}_\ell) + \boldsymbol{\varepsilon}$$

Post-optimality sensitivity update

$$\mathbf{z}' \approx \tilde{\mathbf{z}} - \mathbf{H}^{-1} \mathbf{B} \boldsymbol{\theta}'$$

Sensitivity-based update